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**DS-312 Data Warehousing and Business Intelligence**

**EconReservoir: A Data Lake Comprising of Data  
from Macroeconomics, Microeconomics, Financial  
Markets, Local Markets, and Explainable (News)  
Sources**

**Project Report**

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# 1. Introduction

## 1.1. Project Overview

*EconReservoir* is a comprehensive data lake designed to integrate diverse economic datasets related to Pakistan into a unified, structured, and analytics-ready system. The project consolidates data from multiple domains—including macroeconomic indicators (such as GDP and its components), financial markets (daily stock prices), trade statistics (imports and exports), agricultural outputs, industrial and mineral production, as well as local market price data across major cities. In addition, unstructured and semi-structured sources such as news articles are incorporated to provide contextual and explainable insights.

The system follows a layered data lake architecture to ingest raw data, transform it into clean and consistent formats, and ultimately produce curated datasets that can be easily analyzed.

## 1.2. Domain

This project lies at the intersection of **economics, finance, and data engineering**, with a strong focus on **data warehousing and business intelligence (DW/BI)**. It spans both **macroeconomic analysis** (GDP, national production, trade flows) and **microeconomic/market-level analysis** (commodity prices, stock-level data), enabling a holistic view of the economic landscape.

## 1.3. Objective of the Data Lake

The primary objective of *EconReservoir* is to transform fragmented, heterogeneous, and partially unstructured economic data into a clean, standardized, and integrated data repository. By doing so, the system enables:

- Efficient exploration of relationships and correlations across different economic sectors
- Time-series analysis of key indicators such as GDP, market prices, and production levels
- Integration of qualitative signals (news data) with quantitative metrics for explainable insights
- Support for informed decision-making in areas such as policy analysis, market forecasting, and economic planning

Ultimately, the data lake serves as a foundation for building reliable, analytics-driven insights that can assist stakeholders in understanding trends, identifying patterns, and making data-informed economic decisions.

## 2. Dataset

### 2.1. Sources & Description

#### 2.1.1. GDP Indicators

- **Source:** State Bank of Pakistan
- **Link:** [www.sbp.org.pk/ecodata/index2.asp](http://www.sbp.org.pk/ecodata/index2.asp)
- **License (if any):** Publicly Available
- **Format:** Reports in Excel
- **Role in Data Lake:** Economy of a country has major effects on all its citizens, households, business and markets. Economic growth is a major goal for governments, and people. This is the backbone of the data lake, this may be used to evaluate, how it's effected by some factors, and how it effects the other factors.

#### 2.1.2. PSX Stock prices

- **Source:** Pakistan Stock Exchange Data Portal
- **Link:** [dps.psx.com.pk/historical](http://dps.psx.com.pk/historical)
- **License (if any):** Publicly Available
- **Format:** HTML tables
- **Role in Data Lake:** Financial markets show the true picture of internal trade, investor confidence, and heartbeat of economy. It's effected by production, and other volatile factors often reported in news. Our data lake having a picture of both, may be able to shed some practical predictiveness to these stock prices.

#### 2.1.3. Crop Production

- **Source:** Ministry of National Food Security and Research
- **Link:** [mnfsr.gov.pk/SiteImage/Downloads/Crops%20Area%20AND%20Production%20by%202022-23.pdf](http://mnfsr.gov.pk/SiteImage/Downloads/Crops%20Area%20AND%20Production%20by%202022-23.pdf)
- **License (if any):** Publicly Available
- **Format:** PDF Report with Tables
- **Role in Data Lake:** The core of Pakistan's economy is run by agriculture, it sustains a population of nearly 240 million people, provides a living for a near majority of them. And counts for a major part of our GDP. It's a substantive part of a feedback loop that runs our country (will be discussed later in our report)

#### 2.1.4. Mineral Production

- **Source:** Pakistan Bureau of Statistics

- **Link:** [www.pbs.gov.pk/energy-mining-2/](http://www.pbs.gov.pk/energy-mining-2/)
- **License (if any):** Publicly Available
- **Format:** PDF Report with tables
- **Role in Data Lake:** Production → local markets → financial markets → macroeconomy → microeconomy → local markets → raw material demand → production

### 2.1.5. Industrial Production

- **Source:** Pakistan Bureau of Statistics
- **Link:** [www.pbs.gov.pk/industry-2/](http://www.pbs.gov.pk/industry-2/)
- **License (if any):** Publicly Available
- **Format:** PDF Report with tables
- **Role in Data Lake:** Industrial production drives local market activity, which feeds into financial markets, which in turn shape the broader macroeconomy. Macroeconomic conditions filter back down to microeconomic behavior, consumer spending, business investment, input costs, which influence local market demand, and ultimately determine the appetite and resources for raw material production, closing the cycle.

### 2.1.6. Daily Prices Per District

- **Source:** Pakistan Bureau of Statistics
- **Link:** [www.pbs.gov.pk/price-statistics/](http://www.pbs.gov.pk/price-statistics/)
- **License (if any):** Publicly Available
- **Format:** Excel based annexures
- **Role in Data Lake:** This is an accurate picture of local markets. These effect people's livelihood, and microeconomics for households, and production entities. Hence, not only effecting quality of life, but production levels. Which in turn has a compounding effect on financial markets, and macroeconomics.

### 2.1.7. External Trade (Imports)

- **Source:** Pakistan Bureau of Statistics
- **Link:** [www.pbs.gov.pk/external-trade-statistics/](http://www.pbs.gov.pk/external-trade-statistics/)
- **License (if any):** Publicly Available
- **Format:** Txt file with typewritten reports
- **Role in Data Lake:** External trade provides a clear picture of country's standing with respect to economic consumption vs production and consumer trust.

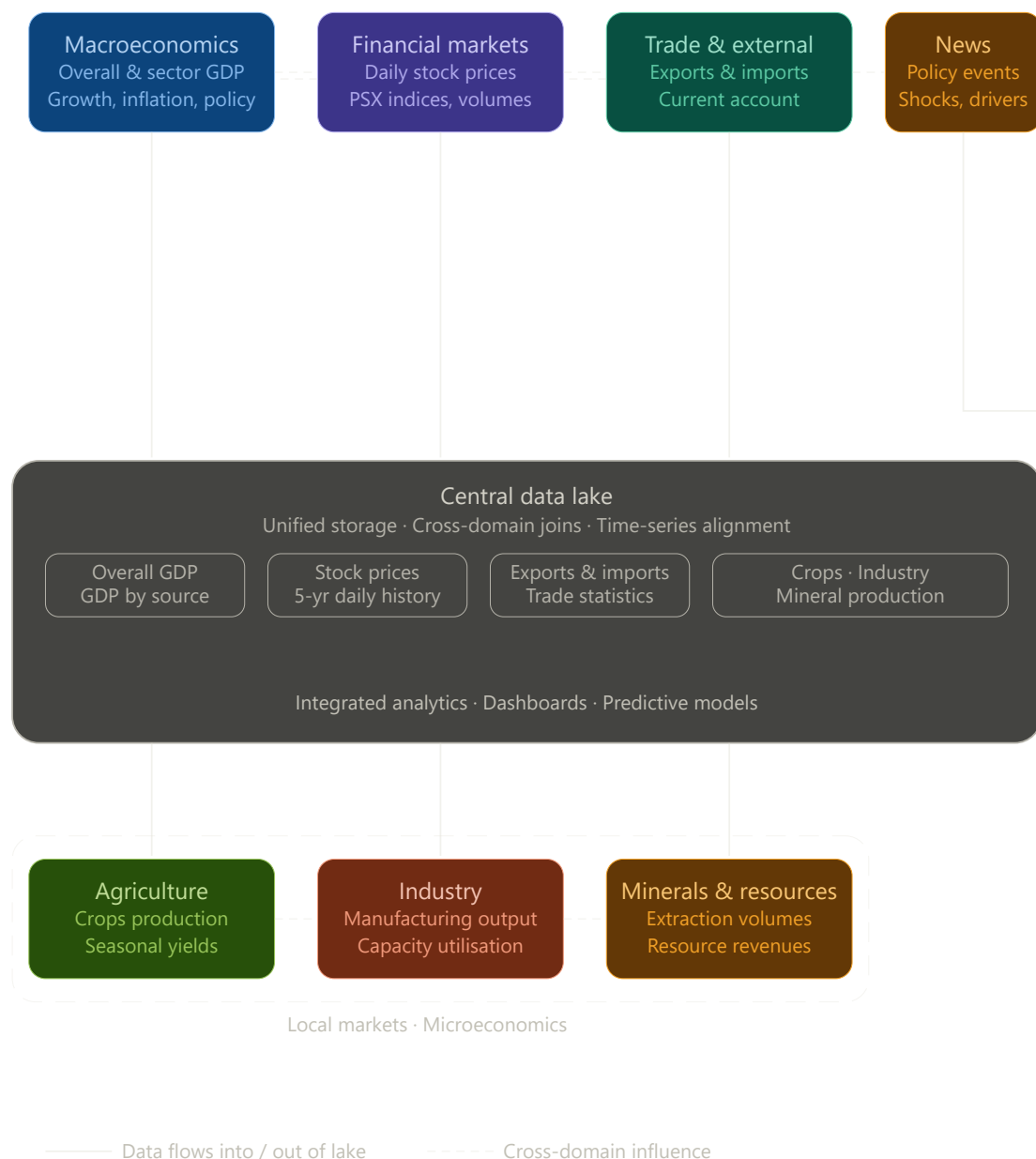
### 2.1.8. External Trade (Exports)

- **Source:** Pakistan Bureau of Statistics
- **Link:** [www.pbs.gov.pk/external-trade-statistics/](http://www.pbs.gov.pk/external-trade-statistics/)
- **License (if any):** Publicly Available
- **Format:** PDF file with typewritten reports
- **Role in Data Lake:** External trade provides a clear picture of country's standing with respect to economic consumption vs production and consumer trust.

### 2.1.9. News Sentiments

- **Source:** Self Curated
- **Link:** Available through email to [mrehan.bsds23seecs@seecs.edu.pk](mailto:mrehan.bsds23seecs@seecs.edu.pk)
- **License (if any):** Does Not Apply.
- **Format:** Text -> ETL + NLP -> CSV
- **Role in Data Lake:** It may provide the necessary reasoning for why an event and occurred and how the public reacted to it. An explainability node in the data lake.

## 2.2. Data Integration and Relation



**Figure 2.1** shows the integration patterns for the selected datasets.

**Top tier (macro + markets + trade + news)** feeds down into the data lake. These are the higher-frequency, nationally reported datasets. GDP and its sectoral breakdown give the headline economic story; stock prices on PSX give you the market's real-time interpretation of that story; trade statistics show the external balance; and news provides the *why*, policy changes, political shocks, weather events, that explains anomalies the numbers alone can't.

**The central data lake** is where cross-domain joins happen. For example, you can ask: "did cotton production this season predict a rise in textile sector stocks and export revenues 2 months later?", a query that spans agriculture, financial markets, and trade simultaneously.

**Bottom tier (agriculture, industry, minerals)** represents the supply-side microeconomic layer. The lake sends enriched signals back down, for instance, using GDP growth forecasts or import data to contextualize whether a drop in mineral extraction is a demand-side or supply-side event.

**Dashed horizontal arrows** show cross-domain influence within each tier, macro-GDP shapes investor sentiment (and vice versa), industrial output drives trade volumes, and crops feed into industry (e.g. agro-processing).



### 3. Methodology

The data lake is constructed using a structured Extract, Transform, Load (ETL) pipeline implemented in Python and PostgreSQL. The pipeline follows the medallion architecture, progressing data through three quality tiers, Bronze, Silver, and Gold, each representing a distinct stage of refinement. The codebase is organized as a collection of domain-specific ETL modules (GDP\_ETL, PSX\_ETL, Crop\_ETL, MINERAL\_ETL, PBS\_ETL, QIM\_ETL, News\_Pipeline) that share a common SQL schema defined in AllQueries.sql

#### 3.1. Extract

Each domain in the data lake has a dedicated Python extraction module responsible for acquiring raw source data and delivering it as flat files ready for ingestion. The seven extraction pipelines cover: GDP sector data from official government statistical releases, PSX historical daily stock prices, commodity price data from the Pakistan Bureau of Statistics (PBS), monthly trade import figures (Table 2 from the Federal Bureau of Revenue), mineral production volumes by province, combined crop area and production data, and news sentiment classifications via the News\_Pipeline module.

The extraction layer does not apply any transformation logic. Its sole responsibility is fidelity, acquiring source data as completely and faithfully as possible, preserving original column names, original string representations of numeric values, and original formatting. This deliberate separation ensures that all subsequent transformations are auditable against the raw source, and that the pipeline can be re-run from scratch without data loss. Each extracted CSV is tagged with metadata fields (`_source_file`, `_ingested_at`) that are carried forward into the Bronze layer to support lineage tracking.

#### 3.2. Transforms: Structuring and Formatting

Before data moves into any storage layer, structural and formatting decisions are made that determine how raw source artefacts will be represented in the database. The most significant of these is the unpivoting of wide-format source tables.

Several source datasets arrive in a wide (crosstab) format where fiscal years or calendar months appear as column headers rather than row values, a common layout in Pakistani government statistical publications. The GDP table, for example, has 25 fiscal year columns spanning 1999-2000 through 2024-25. Trade import data carries twelve-month columns (Jul through Jun). These formats are incompatible with relational storage and time-series querying. The transformation strategy resolves this by converting wide tables to long (tidy) format: each row represents a single observation identified by its entity, period, and value, rather than one row per entity with value spread across many columns.

For the GDP table this means each (`sector_industry`, `fiscal_year`) pair becomes its own row with a single `value_pkr_mn` numeric field. For trade imports, a `CROSS JOIN LATERAL` construct unpivots the twelve-month columns into (`item`, `fiscal_year`, `month_name`, `month_value`) tuples. This structural work happens within the Bronze-to-Silver transition and is codified directly in SQL `INSERT ... SELECT` statements rather than in application code, making the transformation logic transparent and version-controlled.

### 3.3. Load / Bronze Ingestion

The Bronze layer (bronze schema) is the first persistence tier. It consists of seven raw staging tables, `bronze.gdp_table`, `bronze.psx_historical_data`, `bronze.price_data`, `bronze.table2_all_years`, `bronze.mineral_production_all_years`, `bronze.combined_crop_data`, and `bronze.csv1_core`, with a reserved slot for an eighth source.

Every column in every Bronze table is typed as TEXT. This is a deliberate design decision: by accepting all values as strings at ingestion time, the pipeline avoids load failures caused by type mismatches, unexpected formatting in source files, or locale-specific numeric representations. A row that would cause a NUMERIC cast to fail at the Bronze stage is instead ingested successfully and handled, either cleaned or quarantined, at the Silver stage where the rejection rationale can be recorded and audited. In addition to the source columns, each Bronze table carries two system columns: `_ingested_at`, a server-side timestamp populated by `DEFAULT NOW()`, and `_source_file`, a text constant identifying the originating file. These columns form the basis of data lineage from source through to Gold.

### 3.4. Silver Transformations

The Silver layer (silver schema) performs the core data quality work. Each Bronze table has a corresponding Silver table with proper data types, business-meaningful column names, and deduplication constraints. The Silver stage encompasses three categories of operation.

**Type casting and standardisation.** All TEXT columns are cast to their appropriate types, DATE, NUMERIC, INTEGER, or BOOLEAN, using conditional expressions that route invalid values to the reject table rather than raising errors. String values are standardised using `TRIM()` to remove whitespace, `UPPER()` or `INITCAP()` to normalise casing (e.g. stock symbols are uppercased; city and crop names are title-cased), and `REGEXP_REPLACE` to strip extraneous characters (e.g. parentheses in mineral names such as (ONYX) become ONYX). The volume column in the PSX source, which is misnamed and actually contains price change values, is renamed to `price_change` during Silver loading to reflect its true semantic meaning.

**Deduplication.** Each Silver table carries a UNIQUE constraint on its natural key, (trade\_date, symbol) for PSX data, (price\_date, item\_no, city, metric) for price data, (crop, province, district, fiscal\_year) for crop data, and (original\_id) for news sentiment. Duplicate records arriving from Bronze are silently discarded via `ON CONFLICT DO NOTHING`, which ensures idempotency: the pipeline can be rerun without inflating row counts or violating constraints.

**Rejection and quarantine.** A shared `silver.rejected_records` table stores every row that failed validation, along with the source table name, the rejection reason as a human-readable string, and the full original row serialised as JSONB via `ROW_TO_JSON`. Rejection conditions are explicitly defined per dataset: for PSX data, rows with unparseable dates, null close prices, or missing symbols are rejected; for price data, rows with non-numeric values, unknown metric types (MIN/AVG/MAX), or blank city fields are rejected; for crop data, rows with negative area or production figures are rejected; for news sentiment, records with non-integer IDs, malformed dates, or sentiment values outside {Positive, Negative, Neutral} are rejected. A set of post-load data quality checks (DQ-1 through DQ-4) then validates the Silver tables for residual nulls, range violations (e.g. PSX close price  $\leq 0$ , high price less than low

price, GDP growth rates outside  $\pm 30\%$ ), duplicate keys, and provides a tabular reject summary dashboard grouped by source table and rejection reason.

### 3.5. Gold Layer

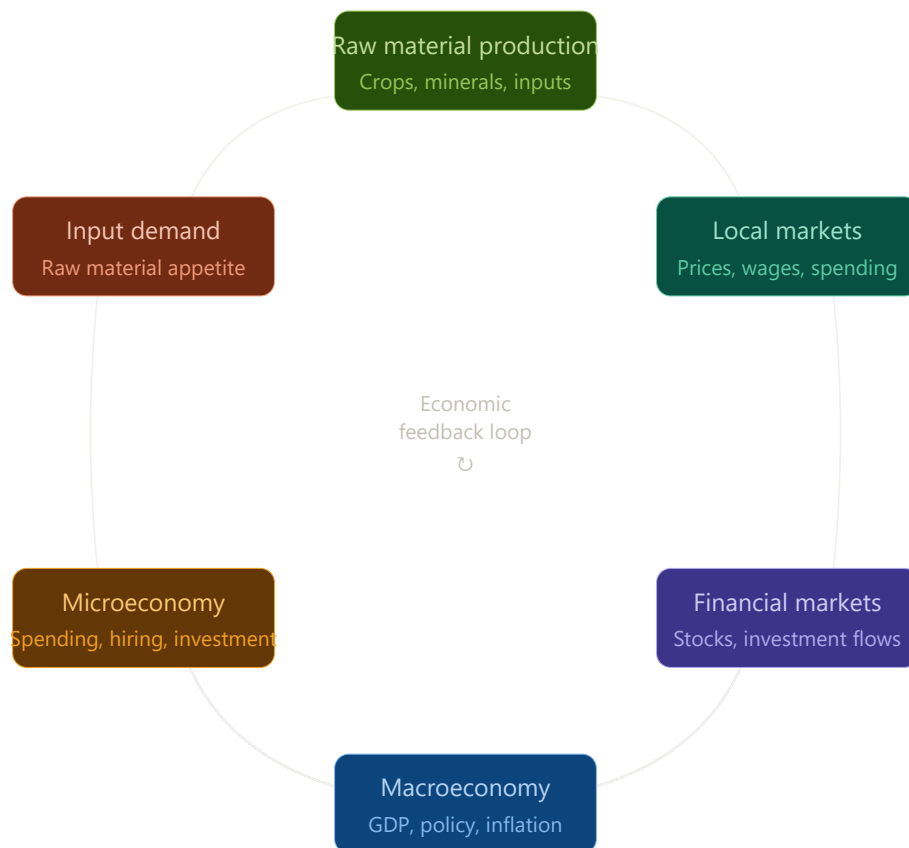
The Gold layer (gold schema) is the consumption tier. It consists of nine curated SQL views derived from the Silver tables, designed to serve analytical queries directly without further preparation. Seven of the nine views correspond to a single domain; the remaining two are cross-domain joins that materialise the relationships at the heart of the data lake's analytical value.

Domain-specific Gold views include: `gold.gdp_sector_growth`, which calculates year-on-year growth percentages for each GDP sector using a LAG window function partitioned by sector and ordered by fiscal year; `gold.psx_daily_summary`, which enriches the OHLC record with computed percentage change, daily price range, and a directional flag (Up/Down/Flat); `gold.price_index_monthly`, which pivots the normalised MIN/AVG/MAX price observations into a monthly city-level index with a derived price spread column; `gold.trade_import_annual`, which reaggregates monthly trade values to annual totals and computes a reconciliation difference against the reported year total as a built-in data quality flag; `gold.mineral_production_summary`, which aggregates monthly extraction volumes by province, mineral, and fiscal year; `gold.crop_performance`, which computes total area, total production, and yield per thousand hectares by province and crop; and `gold.sentiment_dashboard`, which produces a daily category-level sentiment score showing positive, negative, and neutral article counts and percentages.

The two cross-domain views demonstrate the data lake's integrative purpose directly in the schema. `gold.gdp_vs_crop_production` joins the GDP sector growth view to the crop performance view on fiscal year, isolating the agriculture sector GDP alongside Punjab wheat production and yield, making the supply-side to macroeconomic linkage queryable in a single select. `gold.price_sentiment_monthly` joins the monthly price index to the sentiment dashboard on calendar month, enabling correlation analysis between commodity price movements and the tone of news coverage in the same period. These views are the analytical entry points for the cross-domain research described in the knowledge creation section of this report.

## 4. Knowledge Creation & Impact

### 4.1. The Economic Feedback Loop: Core Impact of our Data Lake



**Figure 4.1** shows a constant feedback loop that is said to dictate economic lifecycle.

Economic systems do not move in straight lines. Production, markets, investment, and policy are bound together in a continuous feedback loop, each layer shaping the next, and eventually circling back to influence the conditions from which it emerged. Understanding this loop is the conceptual foundation of this data lake, and it explains why integrating these datasets unlocks a class of insight that no single dataset can provide on its own.

The loop operates across six interconnected stages. Raw material production, crops, minerals, and industrial inputs, determines the cost and availability of goods. Those goods enter local markets, setting wages, employment levels, and consumer spending. Local market performance is then priced into financial markets: investor behavior, stock valuations on the Pakistan Stock Exchange, and capital flows respond to what sectors are producing and selling. These financial signals, in aggregate, shape macroeconomic conditions: GDP growth, inflation, the trade balance, and monetary policy. Macroeconomic conditions in turn feed back into microeconomic decisions, whether a firm expands capacity, whether a household increases consumption, whether a lender extends credit. And those microeconomic decisions determine the demand for raw inputs, completing the cycle and beginning it again.

Each stage of this loop operates on a different timescale. Financial markets react within days. Consumer behavior shifts over weeks to months. GDP figures are reported quarterly. Agricultural output is seasonal. Policy responses lag behind the signals that trigger them. Without a unified data lake that aligns all of these signals on a shared temporal axis, the feedback structure is invisible. Researchers looking at any single dataset see only a fragment, a downstream effect with no upstream cause, or a leading indicator with no lagging outcome to validate it against.

This is the core value of integrating macroeconomic aggregates, sectoral production data, trade statistics, and daily stock price histories into a single queryable resource. It makes the loop legible. A researcher can, for the first time, trace a specific shock, a drought, a currency devaluation, a global commodity price shift, as it propagates forward through the system, observing how it manifests differently at each layer and how long the full transmission takes. That kind of end-to-end visibility is what transforms economic data from a record of the past into a tool for understanding causation.

### 4.2. Stakeholder Identification and Usefulness

The data lake is designed to be a shared research infrastructure, and its value compounds as the range of users who interact with it grows. Different communities of users will approach it with different questions, at different levels of abstraction, and the architecture supports all of them.

Academic economists and data scientists are the most natural primary users. For macroeconomists, the ability to construct high-frequency proxies for GDP growth, using stock market indices and trade flows as leading indicators before official figures are published, represents a significant methodological advance. For development economists, the production-side data on crops, minerals, and industry output provides granular evidence on the structural composition of Pakistani economic growth, enabling analysis of which sectors are driving aggregate performance, and which are being left behind. For financial economists, the five-year daily stock price archive allows systematic analysis of how macroeconomic news events transmit into asset prices, how sectoral shocks are priced in relative to economy-wide shocks, and how market efficiency varies across different information environments.

Policymakers and government institutions represent a second major user group. The State Bank of Pakistan, the Ministry of Finance, and the Planning Commission all make decisions that depend on understanding how different parts of the economy are behaving simultaneously. A model that integrates real-sector data with financial market signals and trade flows gives policymakers a more timely and complete picture than any of those data sources alone. The ability to back-test policy interventions, examining how past rate changes or trade policy shifts propagated through the loop, supports more evidence-based decision-making going forward.

Private sector analysts, asset managers, and research boutiques operating in Pakistani capital markets constitute a third class of users. For these participants, the value lies in the ability to construct systematic, data-driven investment strategies grounded in fundamental economic relationships rather than purely technical signals. Connecting sectoral production volumes to equity performance, or mapping trade statistics to currency dynamics, creates a richer information environment for portfolio construction and risk management.

Finally, civil society organizations, journalists, and policy advocates can use the data lake to hold institutions accountable, tracking whether stated policy objectives are reflected in observable economic outcomes, and whether the benefits of growth are materializing across different sectors of the economy.

### 4.3. Impact on Research

The research implications of this data lake extend well beyond the sum of its individual datasets. By enabling cross-domain analysis at scale, it opens lines of inquiry that were previously either too data-intensive to pursue or simply not possible with available Pakistani economic data.

The most immediate contribution is methodological. Pakistani economic research has historically been constrained by data availability, particularly the absence of high-frequency, machine-readable time series covering both the real and financial sectors simultaneously. This lake substantially relaxes that constraint. Researchers can now construct and test hypotheses about the transmission mechanisms linking production to prices, investment to growth, and trade to domestic output using Pakistan-specific evidence rather than extrapolating from analogous economies.

In the domain of agricultural economics, pairing seasonal crop production data with commodity price movements and trade statistics enables rigorous analysis of how domestic supply shocks pass through into food prices, export earnings, and rural income, questions with direct relevance to food security policy and poverty measurement. In industrial economics, connecting manufacturing output data to stock market performance and import statistics allows investigation of how global supply chain integration affects Pakistani industrial firms and their ability to expand or contract capacity in response to demand signals.

The data lake also enables a new generation of forecasting models for Pakistan. Current approaches to nowcasting Pakistani GDP are limited by the quarterly release cycle of national

accounts data. By training machine learning models on the relationships between high-frequency financial and trade signals and subsequently released GDP figures, researchers can construct more timely estimates of current economic conditions, a capability with both academic and practical value.

At a broader methodological level, the data lake supports the development of Pakistan-specific structural economic models that can be calibrated against observed data rather than imposed from theoretical priors. This matters because the structural relationships that hold in high-income economies, the interest rate channel of monetary policy, the exchange rate pass-through to inflation, the investment multiplier, operate differently in a middle-income economy with a large informal sector, high import dependence, and a volatile external environment. Pakistan-specific calibration, made possible by this integrated dataset, produces models that are both more accurate and more policy-relevant.

### 4.4. Societal Impact

The societal impact of this data lake is ultimately rooted in a simple proposition: better economic understanding leads to better decisions, and better decisions, by governments, businesses, investors, and civil society, produce better outcomes for the people those decisions affect.

Pakistan's economic management faces a persistent challenge that this data lake is well-positioned to help address: the gap between the pace at which economic conditions change and the pace at which those changes become visible to decision-makers. When a drought reduces crop yields, the downstream effects on rural incomes, food prices, and trade balances take months to fully appear in official statistics. By that point, the window for early intervention may have passed. A data infrastructure that makes those effects visible earlier, by tracing how agricultural output signals transmit through local markets and into financial indicators, creates the conditions for more timely and effective responses.

For ordinary Pakistani households, the most direct pathway from this data lake to improved welfare runs through better macroeconomic policy. Inflation in Pakistan has historically been volatile, with food price shocks and currency depreciation disproportionately affecting lower-income households who spend a larger share of income on basic goods. Models that use this integrated data to better predict inflationary pressure, and to distinguish supply-side from demand-side causes, give monetary authorities better tools to respond. Similarly, trade policy informed by granular production and export data can be calibrated to support domestic industries and protect employment rather than relying on aggregate proxies.

At the level of financial markets, improved data infrastructure tends to reduce information asymmetries between sophisticated institutional investors and smaller market participants. When the relationships between economic fundamentals and asset prices are more transparent and better understood, markets price risk more accurately, allocate capital more efficiently, and are less prone to the kind of speculative volatility that erodes the savings of retail investors and pension funds.



Perhaps the most significant long-term societal impact lies in the cumulative effect on research capacity and institutional knowledge within Pakistan. A data lake of this kind, if maintained and expanded over time, creates the conditions for a self-reinforcing research ecosystem: it attracts researchers, those researchers produce findings that inform policy, improved policies generate better economic data, and better data enables more rigorous research. Pakistan has historically faced a deficit of high-quality, locally produced empirical economic research relative to its size and the complexity of its economic challenges. This initiative has the potential to begin closing that gap, not through a single study or policy intervention, but through the sustained accumulation of evidence and analytical capacity that better data infrastructure makes possible.



### 5. Conclusion

EconReservoir demonstrates that Pakistan's fragmented economic data landscape can be unified into a coherent, analytically powerful resource. By integrating nine heterogeneous datasets, spanning GDP, stock markets, trade flows, agricultural output, mineral and industrial production, local commodity prices, and news sentiment, into a single medallion-architected data lake, the project transforms what was previously a collection of isolated snapshots into a living model of the Pakistani economy's feedback loop.

The technical contribution is a reproducible, auditable ETL pipeline that ingests raw government and market data, enforces data quality through structured quarantine and rejection logic, and surfaces curated Gold-layer views ready for immediate analytical use. The intellectual contribution is the demonstration that this integration is not merely convenient but epistemically necessary, the causal relationships between production, markets, macroeconomic conditions, and household welfare are only visible when the data is brought together on a common temporal axis.

EconReservoir is not a finished product but a foundation. Its value will grow as more datasets are added, as the Gold-layer views are extended with machine learning models and forecasting pipelines, and as a broader community of researchers, policymakers, and analysts engages with it. The most important outcome of this project is not any single query or insight it enables today, but the infrastructure it establishes for evidence-based economic understanding in Pakistan for years to come.